

Acidic and Basic

Learning Ladder Assessment Content	7.1.1 Define acids, bases and salts
Lesson duration	40 minutes
Book reference	Science Textbook for Class VII, Chapter 5 Acids, Bases and Salts
Resources	The lesson should ideally be in a science laboratory or other suitable area which allows sufficient space for learners to do practical work in pairs. You will need: • apparatus and solutions for demonstration and practical work: acids and bases, blue and red litmus paper, droppers, tiles, labelled containers of vinegar, baking soda in water, toothpaste in water and aerated drink (e.g. cola) or • appropriate digital resources to demonstrate the concepts

Time (mins)	Learning Outcomes	Lesson Activities	Assessment
	What we want the learners to know, understand and be able to do.		How AfL strategies will be used How AOs will be embedded
10	This lesson focuses on the highlighted part of the following learning outcome: Differentiates materials and organisms such as, digestion in different organisms; unisexual and bisexual flowers; conductors and insulators of heat; acidic, basic and neutral substances; images formed by mirrors and lenses, etc., on the basis of their properties, structure and function.	Start the lesson by asking learners to work in pairs to list five edible things which taste sour. Allow discussion time and then take some responses. Some possible responses are lemon, curd, vinegar, orange juice, raw mango. Ask: Why might these things taste sour? Take some responses. Then explain that these foods contain substances called 'acids' which give the sour taste. Give the success criteria for the lesson. • Understand the terms 'acidic', 'basic' and 'neutral'. • Test if a substance is acidic or basic.	Provide stimulus and use questioning to make connections with existing knowledge Share success criteria for the lesson

10	Show bottles of acids present in the science laboratory (or pictures of them), e.g. hydrochloric acid, sulphuric acid, acetic acid. Explain that laboratory acids can be harmful and should never be tasted. Draw attention to the hazard labels on the bottles and ask: • What does this label mean? • Why is it there? • Is it there on all chemical bottles? • How can acids be harmful? Explain the importance of paying attention to hazard labels and safety in the laboratory. Demonstrate a test to show the effect of acids on blue and red litmus paper. Afterwards, clarify understanding by asking: So how can we test if substances are acidic? Now, introduce the term 'basic' by referring to bases as the opposite of acids and substances that are soapy to touch. Show bottles of bases present in the science laboratory (or pictures of them), e.g. sodium hydroxide, potassium hydroxide, limewater. Also highlight some everyday bases, such as caustic soda and lime. Explain that, like acids, bases can be hazardous and it is important to take care. Demonstrate a test to show the effect of bases on blue and red litmus paper. Afterwards, clarify understanding by asking: So how can we test if substances are basic? Finally, introduce the term 'neutral' to describe substances which are neither acidic nor basic, such as water.	Build new learning by observation and responding to questions Reinforce new concepts by checking through questioning
10	Tell learners that you want them to work in pairs to test four labelled solutions to see if they are acidic, basic or neutral: vinegar, baking soda in water, toothpaste in water and an aerated drink. Ask them to record their predictions for each solution first (for colour changes of red and blue litmus paper and whether they think the solution is acidic, basic or neutral). Learners then do the tests and record their observations. While learners carry out the tests, observe how they handle the apparatus and take safety precautions. Ask learners to compare their observations with predictions and discuss. They then share their observations with the class.	Learners collaborate to apply their learning Learners predict to strengthen understanding and challenge misconceptions

15	Divide the class into groups of five or six. Ask them to discuss: • Where could we use litmus paper testing in daily life? • How could it help us to ensure safety in our surroundings? After five minutes, they join with another group to share their ideas. To conclude the lesson, ask learners to answer the following questions on an 'exit slip' to hand in before they leave the lesson: 1. What happens to the colours of litmus papers if a substance is acidic or basic? 2. What more would you like to know about acidic, basic and neutral substances or what are still confused about? Use the exit slips to inform your next lesson.	Supporting learners in owning their own learning by making real life connections Use exit slips to elicit evidence of learning
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Key competencies

Collaboration

Communication

Critical thinking

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Ask learners to sit very still and count the number of times they breathe in for 30 seconds. Time the 30 seconds for them. Learners then double their value to calculate their resting breathing rate (per minute).

Ask:

1. What do you think would happen to your breathing rate if you exercised?
2. How could you find out which type of exercise has the biggest effect?

Expected responses include:

1. Faster breath, deeper breaths
2. Try different exercises and count breaths, compare with when you are sitting still

As a class, plan a simple investigation to find out which exercise (e.g. walking, jumping, jogging and standing) increases breathing rate the most.

To help emphasise the need to control variables such as length of exercise and resting time between exercises, ask:

1. What will you measure?
2. What will you change?
3. What factors will you control?
4. How long will you exercise for?
5. How will you stay safe?

Responses may include:

1. Number of breaths in 30 seconds
2. Type of exercise
3. How long you exercise, making sure you are breathing at your resting rate before exercising
4. Learners may provide a variety of times, but it is suggested that they exercise for a minimum of 2 minutes in the investigation
5. Choosing where you exercise carefully (away from others, away from furniture, outside of the classroom), wear appropriate footwear, do not exercise if you are not fit and healthy (this may be an issue if they have asthma or similar conditions)

Finalise the plan as a class so that all groups follow the same method.

Support learners in owning their own learning by planning an investigation together

Use questioning to develop scientific thinking